

ZDVINSK, Russian Federation

WMO: 297120

Lat: **54.717N**Lon: **78.667E**Elev: **371**StdP: **14.50**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.7	-26.4	-37.6	0.6	-32.4	-31.6	0.9	-26.2	27.5	13.9	23.3	14.7	3.5	200	0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	85.7	65.8	82.9	65.1	79.8	64.0	69.2	79.9	67.5	78.1	66.1	76.1	8.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	96.0	73.4	63.9	90.2	72.3	62.1	84.6	70.9	33.6	79.9	32.2	78.6	31.1	76.2	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	23.6	20.6	18.3	DB	-36.7	91.2	7.8	3.6	-42.3	93.7	-46.9	95.8	-51.3	97.9	-57.0	100.5
(5)				WB	-36.8	72.1	7.5	2.7	-42.3	74.1	-46.7	75.6	-50.9	77.1	-56.4	79.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.2	-1.3	3.3	18.7	39.3	53.8	63.4	66.8	61.4	51.5	37.8	19.9	5.8
	DBStd	26.22	14.88	13.34	11.56	10.51	9.87	7.34	5.81	6.96	8.42	9.11	13.83	14.44
	HDD50	7047	1592	1307	971	348	78	5	0	3	80	392	902	1369
	HDD65	11117	2057	1727	1436	771	365	113	49	155	414	844	1352	1834
	CDD50	1645	0	0	0	27	196	407	521	357	124	13	0	0
	CDD65	241	0	0	0	0	18	65	106	44	8	0	0	0
	CDH74	2441	0	0	0	7	247	659	939	483	105	1	0	0
	CDH80	614	0	0	0	0	52	161	255	120	26	0	0	0
Wind	WSAvg	8.0	7.8	8.1	9.1	9.0	9.1	7.3	6.4	6.7	7.2	8.0	8.8	8.9
Precipitation	PrecAvg	14.1	0.7	0.5	0.6	0.7	1.0	2.0	2.2	1.9	1.3	1.3	1.1	1.0
	PrecMax	19.0	1.3	0.9	1.6	1.7	3.2	4.0	5.7	4.3	2.9	3.0	2.2	1.7
	PrecMin	8.2	0.1	0.0	0.0	0.1	0.2	0.7	0.4	0.3	0.2	0.4	0.4	0.2
	PrecStd	2.7	0.4	0.2	0.4	0.4	0.6	0.9	1.2	1.0	0.7	0.7	0.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.9	33.5	43.1	73.8	85.2	87.8	90.4	88.2	84.3	68.0	46.8	34.5
		MCWB	31.6	32.1	37.7	55.7	62.9	65.6	68.3	66.2	62.5	52.3	43.2	32.7
	2%	DB	28.6	30.4	38.5	67.0	80.2	84.5	85.8	83.2	76.6	61.4	42.6	32.4
		MCWB	27.5	29.1	34.8	53.5	60.3	65.0	67.3	65.7	59.0	49.8	39.5	31.4
	5%	DB	24.6	26.9	36.0	61.1	76.2	81.2	82.8	79.4	70.9	56.0	38.7	29.2
		MCWB	23.5	25.7	33.4	50.2	59.4	64.2	66.7	63.7	56.5	46.7	36.5	28.2
	10%	DB	21.1	23.0	33.8	55.9	71.4	77.6	79.4	75.6	66.0	50.9	35.6	25.2
		MCWB	20.1	21.8	31.7	47.1	56.2	62.9	65.6	62.3	54.5	44.2	33.9	24.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.9	32.6	37.8	57.9	65.6	70.2	71.8	70.6	64.5	54.1	44.2	33.2
		MCDB	32.5	33.3	41.3	69.9	80.7	82.0	83.0	79.8	77.8	64.0	46.0	33.8
	2%	WB	27.4	29.2	35.5	54.2	62.5	67.7	69.9	67.5	60.0	50.6	39.7	31.4
		MCDB	28.5	30.2	38.0	64.6	77.0	79.6	80.6	78.7	73.4	59.9	42.1	32.4
	5%	WB	23.7	25.7	33.7	50.7	60.1	65.9	68.3	65.7	57.7	47.5	36.7	28.2
		MCDB	24.6	26.8	35.7	60.5	73.7	76.7	78.3	76.2	68.9	54.7	38.7	29.2
	10%	WB	20.0	21.9	31.8	47.3	57.6	64.2	66.9	63.5	55.3	44.8	33.9	24.2
		MCDB	20.9	23.1	33.9	55.2	69.2	74.7	76.5	73.6	64.7	50.5	35.4	25.2
Mean Daily Temperature Range	5% DB	MDBR	14.7	15.7	15.8	17.0	20.8	19.3	18.8	19.9	18.8	14.4	11.2	13.8
		MCDBR	15.2	12.9	13.8	25.5	26.7	25.3	24.7	25.9	27.1	23.4	10.9	13.5
		MCWBR	15.0	12.7	11.5	14.6	12.3	10.5	9.7	11.1	12.7	13.3	8.9	13.3
	5% WB	MCDBR	15.0	13.0	12.3	23.9	24.3	21.4	20.1	22.5	24.4	21.1	10.6	13.8
		MCWBR	15.0	12.9	10.8	14.2	12.6	10.8	9.8	11.5	12.7	13.1	9.0	13.6
Clear-Sky Solar Irradiance	taub	0.291	0.322	0.378	0.406	0.406	0.409	0.436	0.419	0.369	0.353	0.325	0.293	
	taud	2.023	1.981	1.920	2.128	2.215	2.287	2.209	2.229	2.322	2.233	2.077	1.975	
	Ebn at Noon	191	229	242	258	266	266	255	252	250	220	175	156	
	Edh at Noon	23	34	46	43	42	39	42	38	31	26	22	19	
All-Sky Solar Radiation	RadAvg	314	647	1119	1483	1781	1908	1733	1442	1005	514	243	202	
	RadStd	24	36	74	107	152	163	118	117	101	58	38	24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A